

Week 4 Day 4 – Model Tuning, Regularization & Reproducible Pipelines

Tuning Report

Objective

The objective of this project was to build a reproducible machine learning pipeline for predicting whether an individual's annual income exceeds **\$50K** using the Adult Income dataset. The workflow included feature engineering, preprocessing, hyperparameter tuning, model diagnostics, probability calibration, threshold selection, and final model deployment.

Hyperparameter Search

Three models were tuned using **RandomizedSearchCV** with **5-Fold Stratified Cross Validation**. Precision was selected as the optimization metric because reducing false positive predictions was the primary business objective.

Among the evaluated models, **Random Forest** achieved the best performance. The selected hyperparameters were:

- Number of Trees: **200**
- Maximum Depth: **5**
- Minimum Samples per Leaf: **1**
- Maximum Features: **log2**

Model Diagnostics

Learning curves showed that Logistic Regression generalized well with only a small gap between training and validation precision. For Random Forest, increasing tree depth beyond the optimal value increased the gap between training and validation performance, indicating overfitting. Therefore, limiting the maximum tree depth to **5** provided the best balance between model complexity and generalization.

Probability Calibration

The selected Random Forest model was calibrated using **Sigmoid Calibration (CalibratedClassifierCV)**. Calibration quality was evaluated using the **Brier Score** and a calibration plot. Different probability thresholds were also tested, and the threshold was selected based on maximizing **Precision**, which aligns with the business objective.

Final Evaluation

The calibrated model was evaluated on the untouched hold-out test set.

Metric	Value
Accuracy	0.8526
Precision	0.7761
Recall	0.5398
F1 Score	0.6367
ROC AUC	0.9015
PRAUC	0.7724

The evaluation also included a Confusion Matrix, ROC Curve, and Precision-Recall Curve to assess overall classification performance.

Model Deployment

The final calibrated pipeline was saved using **Joblib**. The saved pipeline includes feature engineering, preprocessing, encoding, scaling, probability calibration, and prediction, allowing it to be directly loaded for inference on new data without additional preprocessing.

Conclusion

Random Forest with Sigmoid Calibration was selected as the final model after hyperparameter tuning and model evaluation. The tuned model achieved strong predictive performance while maintaining high precision, making it suitable for production use where minimizing false positive predictions is important.